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SUPERSEDING
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FEDERAL SPECIFICATION

# PLASTIC MOLDING MATERIAL

# PHENOLIC RESIN

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

## 1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers phenolic molding materials which are suitable for compression and transfer molding. The material requirements are derived from ASTM D 700–63T which is an integral part of this specification.

1.2 Classification.

1.2.1 Types. The phenolic molding material shall be furnished in the following types, as specified (see note 1 and 6.2).

Type I —Chemical-resistant materials which are more resistant to deterioration by most commercial chemicals than most phenolic materials. (see note 2).

Type II —General-purpose wood-flour filled materials which are suitable for mechanical and electrical applications.

Type III —General-purpose cellulose-filled materials which are similar to type II but with improved impact strength.

Type IV —Moderate impact strength materials which contain chopped fabrics or other organic fillers (see note 3).

Type V —Medium impact strength materials which contain chopped cotton fabrics or other organic fillers (see note 3).

Type VI —Medium-high impact strength materials which contain cotton fabric, chopped cord, or other organic fillers (see note 3).

Type VII —Low-loss mineral-filled materials formulated for electrical applications.

Type VIII —Superior low-loss mineral-filled materials formulated for electrical applications.

Type IX —Heat-resistant mineral-filled materials for applications that require added mechanical strength.

Type X —General-purpose heat-resistant materials formulated with mineral fillers.

Type XI —General-purpose wood-flour-filled materials designed for mechanical applications (see note 4).

Type XII —Shock-resistant materials which contain paper, flock, or pulp fillers.

Type XIII —Low-gravity heat-resistant materials which contain mineral and other fillers.

Type XIV —Wood-flour-filled rubber-modified phenolic materials formulated for general-purpose applications.

Type XV —Cellulose-filled rubber modified phenolic materials for general-purpose applications.

Type XVI —Mineral-filled rubber modified phenolic materials designed for resistance to heat.

Type XVII —Heat-resistant phenolic materials which contain asbestos fibers for medium-high impact strength (see note 3).



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Type XVIII—Heat-, flame-, and arc-resistant phenolic materials which contain asbestos fibers for medium-high impact strength (see note 3).

Type XIX —Heat- and flame-resistant, high impact strength phenolic materials made with glass fibers (see note 3).

Type XX —Heat-, flame-, and arc-resistant materials which contain mineral fillers for electrical applications.

Type XXI —Superior heat- and flame-resistant materials, with good arc resistance, which contain mineral fillers.

Type XXII —Shock, abrasion, and moisture-resistant materials formulated with synthetic organic flock.

Type XXIII—Low-loss, heat-, flame-, and arc resistant materials which contain short lengths of chopped glass fibers.

Note 1—Molding materials containing either single-stage or two-stage resins are available in most types. Materials containing two-stage resins generally have better storage stability, permit more latitude in molding, and usually have optimum electrical properties. Materials containing single-stage resins have greater freedom from taste and odor and are less likely to cause tarnishing with metals such as brass or silver in enclosures.

Note 2—Chemical-resistant materials may be filled or unfilled. Generally most have a mineral filler.

Note 3—In all cases there will be some sacrifice in appearance and moldability. Transfer molding will result in different physical and electrical characteristics compared to compression molding.

Note 4—Type XXI is similar to type II but has no electrical requirements.

1.2.2 Colors. Phenolic molding material shall be furnished in the following colors, as specified (see 6.2):

Brown
Black
Natural

## 2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP–D–723—Drums, Fiber.

PPP–D–729—Drums; Metal, 55-Gallon (For Shipment of Noncorrosive Material).

Federal Standard:

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following publication forms a part of this specification to the extent



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specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standard:

D 700–63T—Phenolic Molding Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

## 3. REQUIREMENTS

3.1 Material. The molding material shall consist of unmodified or modified phenol-formaldehyde resins which are intimately combined in the uncured or partially cured condition with fillers, pigments, and dyes, as required, to obtain the desired properties. No scrap compound previously cured and re-ground shall be used.

3.2 Property values. Molded specimens of the molding material shall conform to the requirements specified in ASTM D 700–63T. A cross-index of the types in this specification and their ASTM equivalents is shown in table I.

TABLE I. Cross-index of types

| This Specification | ASTM D 700–63T | This Specification | ASTM D 700–63T |
|---|---|---|---|
| Type | Type | Type | Type |
| I | 1 | XIII | 13 |
| II | 2 | XIV | 14 |
| III | 3 | XV | 15 |
| IV | 4 | XVI | 18 |
| V | 5 | XVII | 19 |
| VI | 6 | XVIII | 20 |
| VII | 7 | XIX | 21 |
| VIII | 8 | XX | 22 |
| IX | 9 | XXI | 23 |
| X | 10 | XXII | 24 |
| XI | 11 | XXIII | 25 |
| XII | 12 | | |

3.3 Color. Color and color tolerance shall be as specified (see 1.2.2 and 6.2).

3.4 Workmanship. The molding material shall be uniform in form and composition and shall be free from dirt and other foreign matter.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performing of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Lot. A lot shall consist of all material of the same type produced under similar conditions in one unchanged process by one manufacturer and offered for delivery at one time.



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4.2.2 Lot acceptance tests. Lot acceptance tests shall be made on each lot of material and shall be the basis for acceptance or rejection of the lot. Lot acceptance tests shall consist of the tests specified in 4.5.

4.3 Quality conformance inspection sampling.

4.3.1 Sampling for examination. A random sample of filled containers shall be selected from each lot in accordance with MIL–STD–105, at inspection level I, and the acceptable quality levels (AQL's) specified in table II to verify compliance with all stipulation of this specification regarding requirements not involving tests (see 3.3, 3.4, 5.1, and 5.2).

4.3.2 Sampling for tests. Samples for testing shall be selected at random from each lot. A composite sample, consisting of approximately equal amounts of material obtained from the approximate centers of three containers, shall be used for preparing the required number of molded specimens.

4.4 Examination.

4.4.1 Visual. Examination shall be for the defects and at the AQL's shown in table II.

TABLE II. Examination procedures and standards [1]

| Material | AQL percent | Classification of defect | Defect | Method of inspection |
|---|---|---|---|---|
| Material (see 3.3 and 3.4) | 4.00 | Minor 201 | Improper or nonuniform form [3] | Visual |
| | 4.00 | Minor 202 | Contamination | Visual |
| | 4.00 | Minor 203 | Improper color [3] | Visual |
| Open container [4] (see 5.1) | 4.00 | Minor 204 | Improper or no liner in fiber drum | Visual |
| Closed container [4] (see 5.1 and 5.2) | 1.50 | Major 101 | Exceeds maximum gross weight | Approved scale [2] |
| | 1.00 | Major 102 | Marking misleading or unidentifiable | Visual |
| | 1.00 | Major 103 | Container improperly closed | Visual |

[1] Unless otherwise specified by the procuring agency, MIL–STD–105 shall be used as the basis.
[2] All scales and other measuring tools shall be furnished by the contractor and approved by the Government.
[3] Uniformity of form and (color when required) shall be determined by visual comparison with standards unless other means are required and approved for by the procuring activity.
[4] Applicable only for procurement by the Government.

4.4.2 Packing and marking. Examination shall be for the defects and at the AQL's specified in table II.

4.5 Test methods. Molded specimens shall be conditioned and tested in accordance with the conditioning procedures and test methods specified in ASTM D 700–63T.

## 5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A. Unless otherwise specified, the material shall be packed in one of the following types of containers:

(a) Fiber drums conforming to PPP–D–723, type II, grade A, or type III, grade A in quantities of 200 pounds maximum.

(b) Metal drums conforming to PPP–D–729, type III or type IV, in quantities of 400 pounds maximum.



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Insofar as practical, drums shall be of uniform shape and size, with minimum cube and tare consistent with the protection required. Drums shall contain identical quantities and shall be closed in accordance with the applicable container specification. Fiber drums shall be furnished with a 0.004 inch thick polyethylene liner properly heat sealed.

5.1.2 Level B. Unless otherwise specified, the material shall be packed in one of the following types of containers:

(a) Fiber drums conforming to PPP–D–723, type I, grade A in quantities of 200 pounds maximum.

(b) Metal drums conforming to PPP–D–729, type III or type IV, in quantities of 400 pounds maximum.

Insofar as practical, drums shall be of uniform shape and size, with minimum cube and tare consistent with the protection required. Drums shall contain identical quantities and shall be closed in accordance with the applicable container specification. Fiber drums shall be furnished with a 0.004 inch thick polyethylene liner properly heat sealed.

5.1.3 Level C. Packing shall be in accordance with commercial practice adequate to ensure acceptance and delivery by the carrier for the mode of transportation employed. Containers shall comply with carrier rules and regulations applicable to the mode of transportation.

5.2 Marking.

5.2.1 Civil agencies. In addition to any special marking specified in the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

6. NOTES

6.1 Intended use. The chief characteristics of phenolics are their thermosetting nature, hardness and rigidity, water and chemical resistance, low thermal conductivity, heat resistance up to 250°-475° F., and good electrical insulating properties. Impact strength is dependent upon the type of reinforcing filler used with the resin. Dark colors are recommended since phenolic resins darken with age.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type of material required (see 1.2).
(c) Color required including the tolerances (see 1.2.2 and 3.3).
(d) Selection of applicable level of packing required (see 5.1).

MILITARY INTEREST:
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